THE CO-OPERATIVE UNIVERSITY OF KENYA (CUK)

SCHOOL OF COMPUTING AND MATHEMATICS

LOAN DEFAULT PREDICTION USING CLASSIFICATION TECHNIQUES.

BY

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Project proposal submitted in partial fulfillment of the requirements for the award of the Bachelor of Science in Computer Science.

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DECLARATION AND APPROVAL DECLARATION

DECLARATION

I hereby declare that this Project Loan Default Prediction System Using Classification Techniques is my own work and has, to the best of my knowledge, not been submitted to any other institution of higher learning for any award.

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Signature:		**Date:**	

APPROVAL

This project Loan Default Prediction System Using Classification Techniques has been submitted with my approval as the University supervisor.

Supervisor Name:	**MR. WILSON MUSYOKA**		
Signature:		**Date:**	

ACKNOWLEDGEMENT

I extend my sincere gratitude to my project supervisor, Mr. Wilson Musyoka, for their invaluable guidance, support, and feedback throughout the course of this project. Their expert advice was essential to developing a comprehensive understanding of machine learning techniques for loan default prediction. I am also grateful to my classmates, friends, and family, who provided moral and technical support, and to all individuals and institutions who contributed data or insights, making this study possible.

ABSTRACT

This project explores the development of a machine learning-based model to predict loan defaults, aimed at assisting financial institutions in evaluating applicant risk more accurately. Loan defaults present significant challenges to financial institutions, impacting their profitability and risk management strategies. Traditional risk assessment methods may overlook subtle patterns in data, whereas machine learning models can leverage complex relationships within applicant data to make reliable predictions. This project evaluates several classification models—including Logistic Regression, Decision Tree, Random Forest, and Neural Networks—using financial and demographic features such as income, credit score, and debt-to-income ratio. The chosen model is deployed through a REST API for integration with a mobile application, enabling real-time access for loan officers. The proposed system enhances decision-making, reduces the likelihood of defaults, and streamlines loan processing.

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LIST OF ABBREAVIATIONS

Abbreviation	Definition
ML	Machine Learning
SVM	Support Vector Machine
GBM	Gradient Boosting Machine
API	Application Programming Interface
AUC-ROC	Area Under the Curve – Receiver Operating Curve
SQL	Structured Query Language
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Oversampling Technique
CRISP-DM	Cross-Industry Standard Process for Data Mining
AWS	Amazon web Services

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CHAPTER 1: INTRODUCTION

1.1Background to the Study

Financial institutions face substantial challenges in assessing loan applications accurately due to potential defaults. Loan defaults disrupt profitability and affect the institution's risk portfolio, resulting in heightened scrutiny in lending decisions. Conventional assessment methods rely on manual reviews or rule-based criteria, which may fail to identify complex data patterns associated with loan default risks. Leveraging machine learning, this project proposes a predictive model that classifies applicants based on historical data, empowering financial institutions to make data-driven decisions, reduce defaults, and improve operational efficiency.

1.2Statement of the Problem

Loan defaults present a significant financial risk to lending institutions, affecting their asset quality and profitability. The current methods of applicant risk assessment are often manual, rule-based, and insufficient for identifying subtle patterns in applicant data. Consequently, this project aims to develop a machine learning model to classify applicants based on default risk accurately. Such a model offers an automated, data-driven approach to improve loan approval processes and reduce potential financial losses.

1.3Objectives

1.3.1General Objectives

To develop a machine learning model for predicting loan default risk based on applicant data, enabling financial institutions to make informed lending decisions and reduce financial losses.

1.3.2Specific Objectives

1. To collect and preprocess loan applicant data, including financial and demographic attributes.
2. To train multiple classification models, such as Logistic Regression, Decision Tree, Random Forest, GBM, XGBoost, SVM, and Neural Networks, and evaluate them on accuracy, precision, recall, F1 score, and AUC-ROC metrics.
3. To perform hyperparameter tuning to optimize model performance.
4. To analyze feature importance to identify key predictors of loan default.
5. To deploy the final model via a REST API, enabling integration with a cross-platform mobile application.

1.4 Purpose of the Study

The purpose of this study is to develop an automated, predictive loan default assessment model that helps financial institutions minimize default risks and improve decision-making. The system aims to streamline the risk assessment process, enabling scalability and efficiency in loan processing.

1.5 Significance of the Study

The proposed loan default prediction system benefits financial institutions by:

- Improving Risk Assessment: Identifying high-risk applicants for enhanced loan portfolio management.

- Enabling Data-Driven Decisions: Replacing manual assessments with automated, accurate predictions.

- Enhancing Operational Efficiency: Enabling faster loan approvals and scalability in loan

processing.

1.6 Scope of the Study

This study focuses on building a predictive loan default model using machine learning classification techniques. Data required for the project includes demographic and financial information on applicants, and the study emphasizes model deployment as an API integrated with a mobile app for real-time use by loan officers.

1.7 Assumptions

1. The dataset represents a wide range of loan applicants, capturing diverse profiles.
2. Machine learning techniques and evaluation metrics used will be suitable for loan default prediction.
3. The mobile application will enable effective interaction with the deployed prediction model.

1.8 Limitations and Delimitations

1.8.1 Limitations

1. Data Availability and Quality: Data quality could impact model accuracy.
2. Computational Resources: Advanced models may require significant resources for training.
3. Model Generalization: The model may need further adaptation for different loan types or applicant profiles.

1.8.2 Delimitations

1. This study focuses on supervised classification techniques, excluding unsupervised learning approaches.

2. Deployment is limited to a REST API for mobile app integration, without additional web or desktop interfaces.

1.9 Definition of Terms

Loan Default: Failure of a borrower to fulfill loan payment obligations.

Machine Learning: Algorithm-driven data analysis that enables pattern recognition for predictions.

Classification Model: A model that categorizes data into classes—in this case, default or non-default.

Hyperparameter Tuning: Adjusting model parameters to enhance predictive performance.

Feature Importance: Analysis of features that significantly contribute to model predictions.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Loan default prediction plays a crucial role in helping financial institutions minimize risks and losses. By identifying borrowers likely to default, institutions can make better lending decisions and improve overall portfolio performance. Despite the progress made through traditional credit scoring systems and machine learning-based tools, existing solutions often fail to adapt to emerging challenges. This review explores existing methods, highlights their weaknesses, and discusses how the proposed system aims to address these gaps with a fresh and innovative approach.

2.2 Loan Default Prediction

The of loan default prediction is to use a borrower's financial and personal data to estimate their likelihood of repaying a loan. Traditional systems, such as credit scores, rely on predefined metrics, offering broad evaluations but often missing nuanced patterns. Machine learning has transformed this field by using historical data to detect complex relationships between variables. However, even advanced machine learning models face challenges in practical applications, such as transparency, scalability, and adaptability.

2.3 Existing Systems

Several systems have been implemented to assess loan default risk. These systems demonstrate significant advancements, yet they also have limitations that impact their effectiveness.

2.3.1 Credit Scoring Models (e.g., FICO Scores)

These are rule-based systems that evaluate borrowers based on fixed parameters, such as repayment history and credit utilization. While reliable for basic risk assessments, they lack

the flexibility to incorporate non-traditional data sources or detect subtle patterns in borrower behavior.

2.3.2 AI-based Loan Risk Platforms

Platforms such as Zest AI and other proprietary tools used by financial institutions employ machine learning to predict risk. While they achieve high accuracy, these systems often functions as black boxes, providing little insight into how predictions are made. Additionally, their focus on larger datasets makes them less accessible to smaller financial institutions.

2.3.3 Custom In-house Models

Many banks develop their own prediction systems tailored to their operations. While these models align closely with the institutions' needs, they are resource-intensive to develop and maintain. Moreover, they are often designed for internal use, limiting their scalability.

2.4 Weaknesses of Existing Systems

Existing systems face several common challenges:

1. Static Models: Many systems rely on static data, making them unable to adapt to changing borrower behavior or economic conditions in real time.
2. Limited Features: The scope of features used is often narrow, excluding critical contextual or behavioral indicators, such as spending patterns or localized economic trends.
3. Lack of Transparency: Advanced models, particularly neural networks, lack interpretability, raising concerns about their trustworthiness and regulatory compliance.
4. Scalability Issues: Sophisticated systems require significant computational resources, which are often out of reach for smaller institutions.

5. Imbalanced Datasets: Loan datasets often have a class imbalance, with far fewer default cases than non-defaults. This can skew predictions and reduce model reliability.

2.5 How the Proposed System Will Address These Weaknesses

The proposed system aims to overcome the limitations of existing systems with a unique and practical approach:

1. Dynamic Data Integration
   - ✓ Weakness Addressed: Static models.
   - ✓ Proposed Solution: The system will incorporate real-time data streams, such as credit score updates, transaction patterns, and economic indicators, allowing it to adapt to evolving borrower behaviors and market conditions.
2. Broader Feature Scope
   - ✓ Weakness Addressed: Limited features.
   - ✓ Proposed Solution: Advanced exploratory data analysis will identify and include unconventional data points, such as socio-economic indicators, spending patterns, and borrower financial behaviors, to provide a more comprehensive risk assessment.
3. Explainable Predictions
   - ✓ Weakness Addressed: Lack of transparency.
   - ✓ Proposed Solution: The system will use tools like SHAP (Shapley Additive exPlanations) to explain model outputs, ensuring transparency and building trust among users and regulators.
4. Cloud-based Scalability
   - ✓ Weakness Addressed: Scalability issues.
   - ✓ Proposed Solution: Hosting the system on cloud platforms such AWS or GCP will provide a cost-effective, scalable solution for institutions of all

size.

5. Balanced Predictions with SMOTE
    - ✓ Weakness Addressed: Imbalanced datasets.
    - ✓ Proposed Solution: Techniques like Synthetic Minority Oversampling (SMOTE) will be used to address class imbalance, improving the model's performance on default cases.
6. Cost-effective Accessibility
    - ✓ Weakness Addressed: High resource demands.
    - ✓ Proposed Solution: By designing a lightweight and modular architecture, the system will ensure easy deployment even for smaller institutions with limited computational resources.

2.6 Uniqueness of the Proposed Solution

The proposed system offers a unique value proposition:

- Behavioral-driven Predictions: Including borrower behaviors and socio-economic data ensures that the model provides richer and more accurate insights.
- Hybrid Modeling Approach: Combining classical models (Logistic Regression) with advanced techniques (XGBoost) ensures robustness and adaptability.
- Transparency and Trust: Explainable AI tools will demystify predictions, addressing trust and compliance issue.
- Localized Insights: Contextualizing predictions with regional economic factors ensures relevance across diverse environments.
- Inclusivity: Designed to cater for both large institutions and grassroots lenders, the system ensures accessibility and scalability.

2.7 Conclusion

While existing loan default prediction systems have advanced significantly, their limitations in adaptability, transparency, and scalability leave room for improvement. The proposed system addresses these gaps with innovative methods, ensuring it is dynamic, interpretable, and accessible. This approach not only enhances predictive accuracy but also makes loan assessment more equitable and efficient.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter outlines the planned methodology for building the Loan Default Prediction System. The CRISP-DM framework (Cross-Industry Standard Process for Data Mining) will guide the project. Think of CRISP-DM as a well-planned recipe: each step like understanding the data, cleaning it, and developing the model ensures that the final dish (the predictive system) meets the desired taste (accuracy and reliability). This approach balance structure with flexibility, allowing adjustments at any stage to improve the final outcome. The methodology will involve systematic steps, such as data collection and analysis, preprocessing, model training, evaluation, and deployment.

3.2 Project Design

3.2.1 Development Methodology

The project will adopt the CRISP-DM framework, which resembles the process of designing a car. You start by identifying the purpose of the car (business understanding), select the best materials (data preparation), build the car (modeling), test-drive it (evaluation), finally hand it over to the user (deployment). This framework will ensure every stage is connected and iteratively refined.

The six phases of CRISP-DM are:

- Business Understanding: Define the problem—how can we predict which applicants are likely to default on loans? Establish success criteria, such as improving decision-making for financial institutions.
- Data Understanding: Explore datasets to spot patterns, inconsistencies, and hidden trends. For example, a dataset may show that applicants with higher incomes have fewer defaults, but there might be surprising patterns too, like younger applicants with stable jobs being high-risk defaulters.

- Data Preparation: This involves cleaning messy data, like removing duplicate entries or filling in missing information. It's similar to tidying a mess room—organizing everything to make it usable.
- Modeling: Build machine learning models like Logistic Regression or Random Forest. Each model is like a different tool for the same task; for example, some models excel in identifying patterns in large datasets (Random Forest), while others are simpler but effective (Logistic Regression).
- Evaluation: Evaluate how well the models predict defaults, much like grading a student's performance after a test.
- Deployment: Finally, the best –performing model will be integrated into a system that allows users to input data and receive instant predictions.

3.2.2 Justification for the CRISP-DM Framework

The CRISP-DM framework is ideal for this project as it ensures a deep focus on data preparation and iterative improvement, both of which are critical in building a high-performing machine learning model. Its flexibility allows adjustments in earlier phases if new insights arise during later stages. Moreover, its emphasis on connecting technical work with business needs ensure that the project remains user-focused.

3.3 Design Procedures

3.3.1 Requirement Analysis:

At this stage, we'll create a checklist of what the system needs to do. Think of it as planning the blueprint for a house. Key questions include:

- What data will the system process? (Loan applications, financial details).
- What should the system deliver? (A yes/no prediction of loan default, along with confidence scores)

- What constraints are there? (The system must be scalable, secure, and efficient).

3.3.2 System Architecture:

The system will have a client-server architecture. Imagine a restaurant kitchen where chefs (the server) prepare meals (predictions) based on orders (input data) from customers (users). The chef works behind the scenes, ensuring quality, while the customer only sees the final dish.

3.3.3 Workflow Design:

The workflow will be modular, allowing each component to be fined-tuned without disrupting the rest. For instance:

- Data cleaning will be handled by a preprocessing module.
- Model training will happen separately so that new models can be added later if needed.
- Evaluation: Comparing models using well-defined metrics.

3.4 System Requirements

3.4.1 Functional Requirements

The system will:

- Import raw datasets (financial and demographic data).
- Clean, preprocess, and transform the data into a machine-learning-ready format.
- Train and, test, and evaluate multiple machine learning models.
- Predict the likelihood of loan default for new applicants.

- Allow secure data access through API for authorized users.
- Log and store prediction results securely for auditing or future improvements.

3.4.2 Non-Functional Requirements

- Security: All data communication will use encryption (HTTPS). Imagine a lockbox where sensitive information is stored, and only those with key (authorized users) can access it.
- Reliability: The system will guarantee uptime and provide consistent results. It will aim 99.9% uptime to provide consistent predictions.
- Scalability: As the number of users grows, the system should be able to handle more requests without slowing down.
- Performance: Predictions will be generated in under one second, ensuring real-time usability.

3.4.3 Software Requirements

The system will rely on:

1. Programming Tools: Python, Flask/Django for API development.
2. Data Analysis Libraries: pandas, Numpy, scikit-learn, TensorFlow, and XGBoost.
3. Database: MySQL or PostgreSQL for storing prediction logs.
4. Hosting Platform: Docker containers for portability and AWS EC2 for deployment.
5. Visualization Tools: Seaborn, Matplotlib, and Plotly for generating analytical insights.
6. Development Tools: Jupyter Notebook, Anaconda, Notepad ++, VisualStudio Code.

3.4.4 Hardware Requirements

For development:

- Processor: Intel core i7 or equivalent.
- RAM: Minimum 8GB for handling large datasets.
- Enough storage especially SSD for fast read/write operations.

For deployment:

- A cloud server with at least 4 CPUs, 16 GB RAM, and scalable storage to host the predictive model.

3.5 Data Collection and Analysis

3.5.1 Data Collection

Data will be collected from two sources:

- Public Datasets (Secondary Data): Public datasets, like those available on Kaggle or UCL Machine Learning Repository, will be used as secondary data sources. These datasets often contain large volumes of financial and demographic information that have been collected for research or business purposes. For example, datasets might include details on loan applicants such as income, credit history, loan amounts, and repayment histories. By using these secondary data sources, the project can save time on data collection and focus on building predictive models.
- Synthetic Data: In addition to secondary data, synthetic data will be generated to ensure privacy compliance, as using real-world financial data might raise privacy concerns. This synthetic data will mimic the patterns of real applicants without revealing personal details.

3.5.2 Data Analysis

Once collected, the data will undergo a thorough analysis process:

- Univariate Analysis: Analyzing individual variables, such as the distribution of income or loan amounts in the secondary dataset, will give insights into trends and patterns among applicants.
- Bivariate Analysis: Investigating relationships between key features in the secondary datasets, such as how income correlates with loan repayment behavior, will be crucial for building the predictive models.
- Correlation Analysis: Identifying features that are highly correlated will help to streamline the model by removing redundant features, ensuring better performance and efficiency.

3.5.3 Data Preprocessing

Data preprocessing will be done to prepare the secondary data for modeling. Steps will include:

- Handling Missing Data: Missing values in the secondary data, such as incomplete credit histories or missing loan repayment records, will be addressed by imputing values based on available data or trends.
- Normalization: Features such as loan amounts and applicant ages from secondary data will be normalized to ensure they are on the same scale, which is critical for machine learning algorithms.
- Outlier Detection: Any outliers in the secondary data, such as applicants with very high or low income, will be identified and dealt with to prevent model distortion.

3.6 Model Training and Evaluation

3.6.1 Algorithms to Be Used

Different algorithms will be tested to find the best fit for this problem, such as:

- Logistic Regression for its simplicity and interpretability.
- Random Forest for handling large datasets and identifying important features
- XGBoost for its efficiency in predicting complex relationships.
- Gradient Boosting Machines (GBM).

3.6.2 Evaluation Metrics

We'll evaluate models using metrics like:

- Accuracy: How often the model predicts correctly.
- Precision and Recall: These help assess performance on positive cases (e.g., correctly predicting defaults). To measure the trade-off between false positives and false negatives.
- F1 Score: A balance between precision and recall.
- AUC-ROC: A score that shows how well the model separates defaulters from non-defaulters.

3.7 Deployment

The final model will be deployed as a REST API hosted on AWS. Think of it as setting up a vending machine—users can interact with it through an interface, but all the heavy lifting (like training models) happens behind the scenes.

3.8 Ethical Considerations

- Data Privacy: All personal information will be anonymized before use.
- Bias Mitigation: Diverse datasets will be used to ensure fairness in predictions.
- Explainability: Tools like SHAP will provide insights into why the model predicts

default or no default, helping end users build trust in the system.

3.9 Conclusion

This Methodology ensures a practical, structured approach to solving the problem of loan default prediction. By iteratively refining each stage and aligning it with real-world business needs, the project will result in a scalable, reliable, and transparent system that financial institutions can trust.

APPENDICES

Table 1: Project schedule

Task	**Duration(weeks)**
Conceptualization and Scoping of Project Title	1
Proposal writing and submission to DCIT	2
System Development	4
Project Report writing	5
Project Presentation and Submission	1
Corrections and final submission	1

Table 2: Project budget

S/N	**Items**	**Specification**	**Quantity**	**@ Ksh**	**Amount (Ksh)**
1	Printing paper	A4, High quality	2 reams	500	1,000
2	Internet	5 GB	4 months	1,500	6,000
3	Transport				2,200

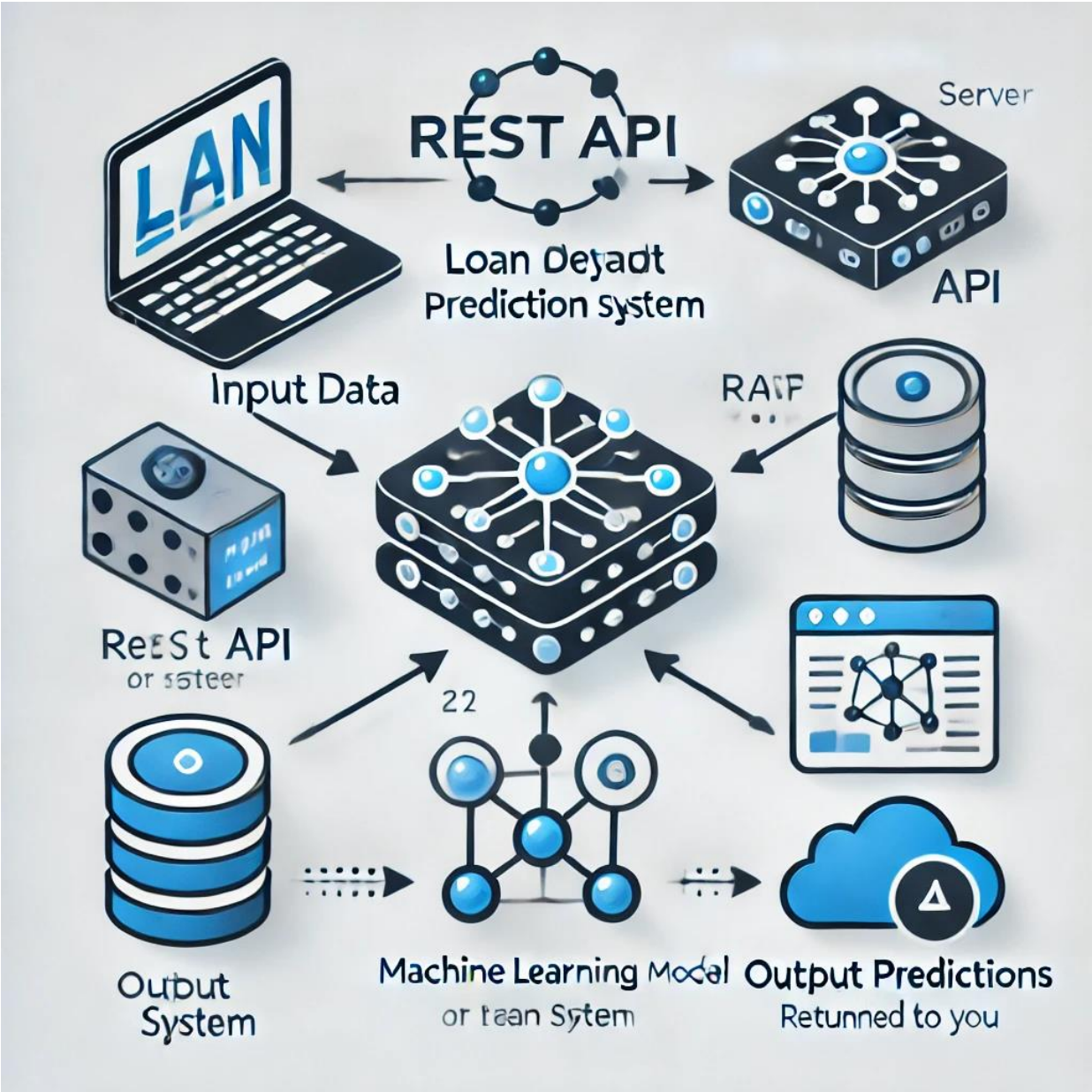
Table 3: Gantt chart

	Duration(weeks)													
Item of work	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Conceptualization and Scoping of Project Title														
Proposal writing and submission to DCIT														
System Development														
Project Report writing														
Project Presentation and Submission														
Corrections and final submission														

Table 4: Evaluation Metrics

Metric	Description	Purpose
Accuracy	Overall correctness of the model	Measures general model performance
Precision	Proportion of correctly predicted defaults	Focuses on minimizing false positives
Recall	Proportion of actual defaults correctly predicted	Focuses on minimizing false negatives
F1 Score	Harmonic mean of precision and recall	Balances precision and recall
AUC-ROC	Area under the Receiver Operating Characteristic Curve	Evaluates trade-offs between sensitivity and specificity

Figure 1: Deployment Architecture for Loan Default Prediction



REFERENCES

1. Brownlee, J. (2020). *Master Machine Learning Algorithms*. Machine Learning Mastery.
2. Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research.
3. Shapley, L. (1953). *A Value for n-Person Games*. RAND Corporation.
4. Kaggle. Loan Default Dataset. Retrieved from https://www.kaggle.com/.
5. Zhang, C., & Ma, Y. (2012). *Ensemble Machine Learning: Methods and Applications*. Springer.
6. ZestFinance. (2023). AI-based Credit Risk Prediction Systems. Retrieved from https://www.zest.ai/.
7. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.